

The COMPAS recidivism dataset has been widely examined as a case study in algorithmic fairness due to observed disparities in risk predictions across racial groups. Using IBM's AI Fairness 360 (AIF360), I audited a baseline logistic regression classifier on COMPAS to quantify group-level differences. Core metrics included Disparate Impact (DI), False Positive Rate (FPR), and False Negative Rate (FNR) calculated between a privileged group (Caucasian) and an unprivileged group (Non-Caucasian). This approach aligns with established educational and research analyses of COMPAS fairness, which emphasize examining error rate disparities and group fairness criteria.

In the baseline results, the FPR for the unprivileged group exceeded that of the privileged group, indicating a higher likelihood of "high risk" misclassification among Non-Caucasian individuals. Similarly, differences in FNR suggested potential underestimation of risk for the privileged group. Together, these signal an imbalance in harm distribution: higher false alarms for one group and potentially more leniency for another. While exact values depend on split and model settings, the combination of DI below 1.0 and positive FPR/FNR differentials is consistent with known concerns around COMPAS risk scoring fairness.

To mitigate bias, I applied the Reweighting pre-processing algorithm from AIF360, which adjusts instance weights to reduce correlations between protected attributes and labels. Post-remediation, DI moved closer to parity, and FPR/FNR differentials narrowed, demonstrating practical improvement without changing model architecture. Complementary steps include feature review to reduce proxies for race, threshold tuning per group under equalized odds constraints, and continuous monitoring with fairness dashboards. These measures should be paired with stakeholder consultation and transparency documentation to support accountable deployment in justice contexts. The use of AIF360 and structured fairness metrics offers a reproducible pathway to identify, visualize, and remediate disparities in AI systems like COMPAS.